

“THIN-FILM MODULES REDUCE THE COST OF ENERGY GENERATION”



■ **DR RANDHIR THAKUR**
SENIOR VP AND GM OF
APPLIED'S SunFab THIN
FILM SOLAR AND DISPLAY
BUSINESS GROUP

Applied Materials, Inc. (AMAT) is the global leader in nanomanufacturing technology solutions with a broad portfolio of innovative equipment, services and software products for fabrication of semiconductor chips, flat-panel displays, solar photovoltaic cells, flexible electronics and energy-efficient glass.

Dr. Randhir Thakur, senior vice president and general manager of Applied's SunFab Thin Film Solar and Display Business Group, was recently in India to announce that Moser Baer India is their first Indian customer that will start volume production of thin-film photovoltaic (PV) modules. On this occasion, Dr. Thakur, along with Rajiv Arya, CEO of Moser Baer Photovoltaic (MBPV), spoke to EFY's Uma Bansal about the solar energy market and Applied Material's advanced SunFab solar technology as a cost-effective option for clean renewable energy



■ **RAJIV ARYA**
CEO OF MOSER BAER
PHOTOVOLTAIC (MBPV)

What is the current state of the solar energy technology market in India and globally?

RT: According to Solar Buzz, annual growth of the global photovoltaic (PV) market has averaged a compound rate of just over 47 per cent since 2003 and strong growth is expected into the foreseeable future. In 2007, world PV cell production was 3036 MW for crystalline silicon and 400 MW for thin-film. Thin-film cell production now represents 12 per cent of global PV cell production and this percentage is expected to increase.

India is regarded as one of Asia's fastest growing economic powerhouses and, as a result of this growth, there is increased demand for electricity. As a matter of fact, India is the world's third largest consumer of coal, which accounts for more than half of India's total energy generation. However, even the current consumption of coal is not enough to meet India's demands. Of the estimated 132 GW of energy capacity in India, only 110 MW is derived from solar.

Therefore the increased demand for electricity, coupled with the dwindling supply of fossil fuels, represents significant opportunities for all sources of renewable energy, especially solar.

What are the benefits of thin-film PV over crystalline silicon cells?

RT: Each technology is well positioned to serve a unique market segment. For space-constrained areas such as residential and small commercial rooftops, crystalline silicon panels are ideal because of their higher efficiency.

For large commercial and utility-scale solar farms, thin-film offers tremendous cost-saving benefits that will help drive the cost of solar power generation toward grid parity

and beyond. Thin-film also has some technical advantages for these types of deployments. For example, when a solar cell is shaded, the output of the entire module decreases. With the unique size of the SunFab produced panels, the cell is as long as the module—up to 2.6 metres in length. This makes it very difficult for any shading to cover the entire cell to the extent that it decreases power production. Additionally, solar panels overall can lose power output when they reach extremely high temperature, like in very sunny places on hot summer days. Thin-film modules retain more of their power output than crystalline silicon under these conditions because of differences in the materials that make the cell.

Which products use the thin-film PVs and what are the market opportunities for these products?

RT: Any product can use thin-film PVs. From applications as small as calculators to utility-scale installations and building integrated photovoltaics, the market opportunity is virtually endless.

How does Applied Materials' technology benefit Indian customers?

RA: The thin-film line that we have established with Applied Materials is a fully-integrated PV module production line that delivers leading-edge solar manufacturing capability using advanced engineering, process equipment, automation and other key supporting technologies. The technology will provide innovative PV products that can significantly reduce electric power and utility costs and provide a cost-effective option for clean and renewable energy.

Thin-film modules can facilitate reduction in the cost of energy generation for two reasons: thin-film amorphous silicon is more appropriate for warm climates and large